Chemistry Lab

Division C

Test Document

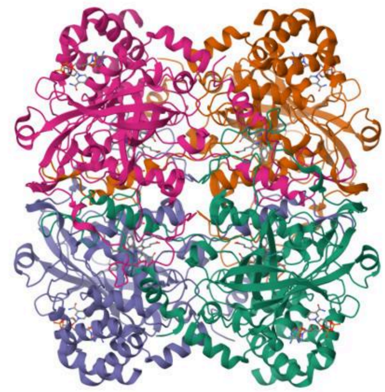
2025 University of Chicago Science Olympiad Invitational

****Please do not write on the test****

Amer Dzankovic (UChicago'26)

Section I: Lab Experiments **(50 points)**

Experiment 1: Catalase Function **(15 points)**



Biological processes often depend on chemical reactions that occur spontaneously but over time periods that would jeopardize the sustainability of life. To ensure these reactions occur within a reasonable timepoint inside the organism, enzymes assist with these processes as catalysts. Catalase is one such important enzyme that helps with the breakdown of hydrogen peroxide (H_2O_2) into water and oxygen. Hydrogen peroxide is a toxic product of normal metabolic activity that in excessive amounts can induce oxidative damage to cellular components like DNA. Catalase is a powerful antioxidant enzyme that ensures this reaction occurs in a timely manner, preventing oxidative stress buildup in the body, and is found in high concentrations in the liver.

FigureRCSB PDB: *Human Catalase* Fan, H.C., Zhang, Y., Sun, F.

Materials

- 2 test tubes
- 2 small samples of chicken liver (~2g) (catalase source)
- Warm Water Bath (at front of classroom)
- Ice Bath (at front of classroom)
- 3% Hydrogen Peroxide

Procedure

1. Fill each glass test tube with a small amount of liver, about 0.5 cm worth.
2. Place one test tube in the warm water bath and the other in an ice water bath
3. Set a timer for 5 minutes
4. Place 2 mL of 3% hydrogen peroxide in each tube AFTER the incubation periods are over and note your observations.

Questions

1. Write the balanced chemical reaction for the reaction catalyzed by catalase
2. Note your observations when adding hydrogen peroxide to the solution of liver
3. What effect does temperature have on catalase activity? Do you think a boiling water bath would affect the reaction rate? How so?
4. What do you think would happen to the reaction if we added 10% Hydrogen peroxide instead of the 3%?

Experiment 2: Iodine Clock Reaction **(35 points)**

In this experiment, you will explore a common reaction used to observe the effects of initial reactant concentration on chemical kinetics: the iodine clock reaction. In this experimental setup, we will rely on a couple of "competing" reactions that occur to give us different reaction times for the clock reaction.

The primary reaction to be studied is the oxidation of I^- by $S_2O_8^{2-}$ (persulfate) in aqueous solution, presented in equation A. This reaction, in the presence of starch, allows I_2 to form a blue-black iodine-starch complex. However, a second reaction presented in Equation B, shows how $S_2O_3^{2-}$ (thiosulfate) ions can "quench", or use up, the I_2 produced in Equation A to prevent it from reacting with starch and forming the sudden color change we expect to see.

$$\mathbf{2I^-(aq) + S_2O_8^{2-}(aq) \rightarrow I_2(aq) + 2SO_4^{2-}(aq)} \quad \text{(slow, rate determining)} \qquad \textbf{Equation A}$$

$$\mathbf{I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)} \quad \text{(fast)} \qquad \textbf{Equation B}$$

$$\mathbf{I_2 + starch \rightarrow color\ change} \text{ (clock reaction)}$$

We will run parallel reactions with a known amount of $S_2O_3^{2-}$ in each run. Once all of the $S_2O_3^{2-}$ ions have reacted, any further I_2 formed by equation A will react with starch to initiate the clock reaction. We will observe the effect of reactant concentration on rate of reaction by measuring the time it takes for the clock reaction to initiate.

NOTE: Any excess I_2 that does not react is TOXIC, and can be an eye and skin irritant. Thus, it is **required** to wear the proper PPE to conduct this experiment.

Materials

- 0.2% starch solution
- 0.012 M $Na_2S_2O_3$
- 0.20 M KI
- 0.20 M KNO_3
- 0.20 M $(NH_4)_2S_2O_8$
- 0.20 M $(NH_4)_2SO_4$
- 2 erlenmeyer flasks (One labeled A one labeled B)
- Waste Beaker
- SAFETY EQUIPMENT (GOGGLES, GLOVES, LAB COAT)

Procedure

1. Label one erlenmeyer flask (beaker) A and the other erlenmeyer B
2. For steps 3-4 refer to CHART A below
3. In **flask A**, place the correct amount of **$Na_2S_2O_3$, KI, KNO_3 , and starch** listed for Run 1. Swirl.
4. In **flask B**, place the correct amount of **$(NH_4)_2S_2O_8$ and $(NH_4)_2SO_4$** listed for Run 1. Swirl
5. Have one partner pour the contents of flask B into flask A. Have another partner start a stopwatch as soon as the contents are poured into flask A. Gently swirl the flask.
6. STOP the stopwatch when the flask turns blue black (the iodine reaction has completed and the starch complex has formed at this point)
7. Empty the contents into the waste beaker at your bench, and rinse the flask with water from a wash bottle.
8. Conduct two more trials for Run 1 to obtain two more reaction times.
9. Repeat Steps 1-8 for the other Run numbers using the corresponding volumes in CHART A.
10. Fill out your data in CHART B and calculate the rate, average rate, and k value.

CHART A

	Beaker A				Beaker B	
Run No.	**Starch Drops**	**0.012 M $Na_2S_2O_3$ (mL)**	**0.20 M KI (mL)**	**0.20 M KNO_3 (mL)**	**0.20 M $(NH_4)_2S_2O_8$ (mL)**	**0.20 M $(NH_4)_2SO_4$ (mL)**
1	5	1	4	1	2	2
2	5	1	2	3	2	2
3	5	1	1	4	2	2

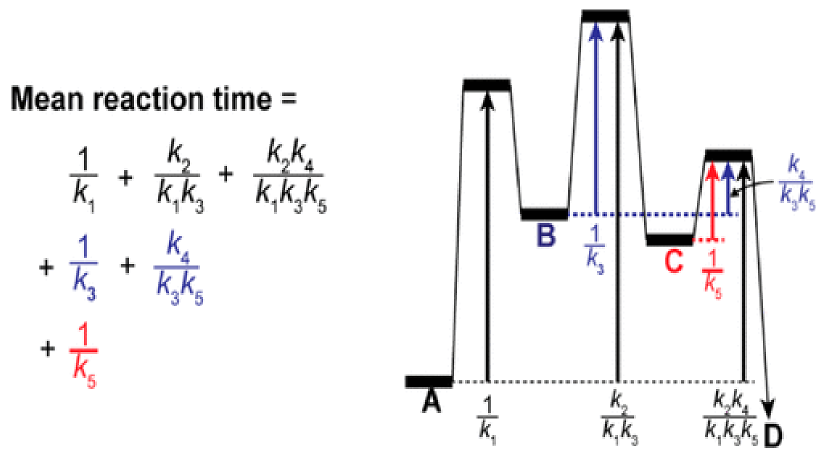
CHART B

Run No.	Initial [I-] ( )	Initial $[S_2O_8^{2-}]$ ( )	time ( )	rate ( )	Average Rate ( )	Average Rate Constant (k) ( )
Run 1a						
Run 1b						
Run 1c						
Run 2a						
Run 2b						
Run 2c						
Run 3a						
Run 3b						
Run 3c						

Section II: SAQ

Chemical Kinetics for Mathematical Theorists

1. A recent analysis conducted by Chiwook Park derives a new term coined the "mean reaction time" **(τrxn)**, or the average time required for the completion of a chemical reaction[1]. The formula for τrxn is derived based on basic chemical kinetic models learned in a high-school chemistry course where each successive K constant either represents the forward or backward portions of reversible reaction. The following questions relate to Park's findings and aim to analyze them from a chemical perspective.



a. The above image is one of the main figures used by Park to represent the different components of his Mean reaction time equation. Please label the following on the figure: **activation energies, time, Gibbs Free energy, intermediates. (5 points)**

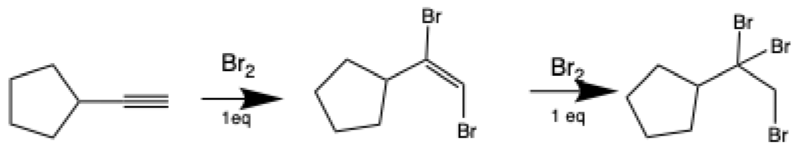
b. Based on the figure above, can we tell whether the overall reaction, A → D is overall exothermic or endothermic? How do you know? Does the calculation of **τrxn** depend on this? **(5 points)**

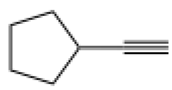
c. You are hired as an undergraduate grader at UChicago's Chemistry department after taking Honors General Chemistry I. A student has submitted a regrade request for a question they believe they have answered correctly. The question and the student's response are both listed below. Please provide a response to the grading inquiry and either agree or disagree with the student, citing evidence as to why your grading is correct or incorrect. **(10 points)**

Question: "Please explain which three components of Park's diagram must be added to obtain the overall activation energy"

Student Answer: $\frac{1}{K_1} + \frac{K_2}{K_1+K_3} + \frac{K_2K_4}{K_1\,K_3K_5}$

d. A direct statement by Park in the paper reads **"For higher-order reactions, the concept of the mean reaction time is not applicable because the reaction time of one reactant is dependent on the concentrations of other reactants and the concentrations of other reactants vary during the reactions."** Based on this statement, will you be able to compute and present a **(τrxn)** for the halogenation of an alkyne (triple bond) based on your experimental results below? **(5 points)**



Trial	Concentration of 	Concentration of Br_2	Reaction Rate
1	0.068 M	0.110 M	0.248
2	0.069 M	0.225 M	0.496
3	0.180 M	0.200 M	1.488
4	0.137 M	0.111 M	0.497

e. Given the following scheme from the paper, and the information in the reaction mechanism above, why can't a mean reaction time be calculated for the bromination of an alkyne? What assumptions are made for Park's calculations? **(3 points)**

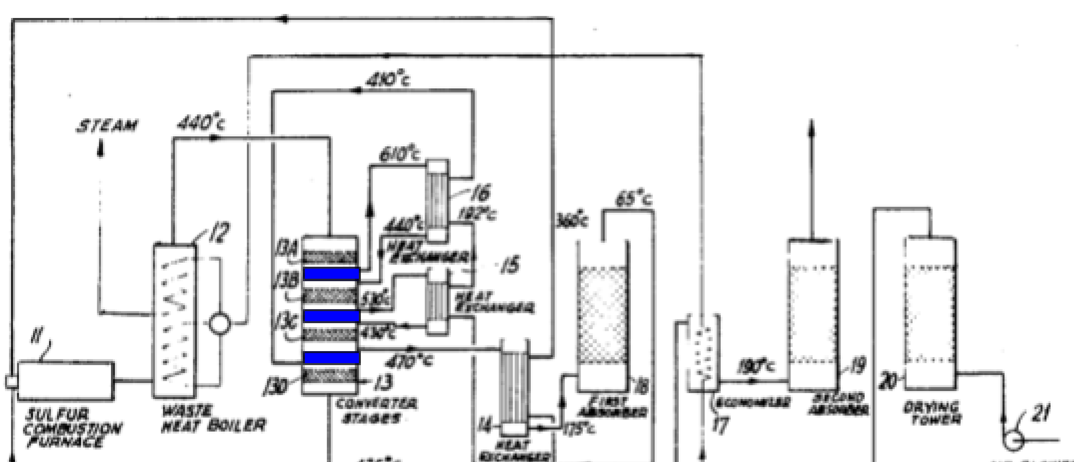
$$A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B \underset{k_4}{\overset{k_3}{\rightleftharpoons}} C$$

2. Sulfuric Acid is commonly used for fertilizer production, petroleum refining,metal processing, and even acts as the electrolyte in lead-acid batteries. The contact process is an industrial method for the production of sulfuric acid via a vanadium(V) oxide catalyst at high temperatures of 450ºC.

a. Please write out the balanced chemical reactions involved with the following individual steps used in industrial sulfuric acid production. **(8 points)**

I. Molten (solid) sulfur is burned in air to produce sulfur dioxide
II. Sulfur dioxide is added into an excess of oxygen in the presence of a vanadium pentoxide at 450ºC and 1-2 atm to make sulfur trioxide
III. Sulfur trioxide formed is added to sulfuric acid to give rise to oleum (disulfuric acid)
IV. Oleum is then added to water to form very concentrated sulfuric acid

b. An engineering intern has been tasked with optimization of the sulfuric acid yield measured at the end of the oleum mixing tank. He proposed the addition of three new catalyst beds in the catalytic converter, colored blue below, that will fit between existing catalyst beds and have the same dimensions as existing ones. Another intern argues that this will only introduce increased hassle since the catalytic beds probably need to be replaced after their consumption in the reactive process. Is this a design that should be invested in? Why or why not? **(5 points)**



United States Patent, 1971
Walter Jaeger Stroudsburg, Pa.

c. Write the chemical formulas associated with the following two reactions involving part of the catalyst beds.

I. Sulfur Dioxide is oxidized into Sulfur Trioxide using Vanadium (V) **(2 points)**
II. Vanadium(IV) is turned back into Vanadium (V) by reacting with dioxygen **(2 points)**

d. The following excerpt is taken from the conclusion of the patent (lines 30-40)

sistant alloys. In addition, a process is provided in which the final effluent gas leaving the last absorber is sufficiently clean of sulfur to meet newly imposed legislative standards. For example, a feed gas containing 11 percent sulfur dioxide can be substantially converted to sulfur trioxide with a yield of 99.5 to 99.8 percent so that the amount of sulfur dioxide remaining in the tail gas may range from approximately 0.02 to 0.05 percent or range up to about 500 parts per million parts of gas which is substantially below the amounts (e.g. 3,000 to 5,000 parts per million) which prevail in the more conventional systems.

If 1 L of sample gas at standard temperature and pressure were taken from the conventional system containing 3000 parts per million sulfur dioxide, how many molecules of sulfur dioxide gas would be contained within the sample? What is this value for the maximum 500 parts per million collected in the new system designed in the patent? **(5 points)**

e. The engineering intern reads further through the patent and finds the equation below modeling the surface area of the heat exchanger system. You are to propose the most efficient copper alloy to use to minimize the surface area of the what exchanger system and thus the cost of construction for the site. Using the equation and the list of copper alloys, which copper alloy would minimize the surface area of the heat exchanger system? **(5 points)**

$$A=\frac{Q}{K\times\Delta Tm} \tag{1}$$

where

A is the surface area of the heat exchanger system
Q is the heat involved in the heat exchanger system
K is the overall heat transfer coefficient expressible as B.t.u./hour/sq. ft/° F. or as Kcal./hr./M²/°C.
ΔTm is the log mean temperature difference. The mean of the terminal temperature difference, ΔTm, is defined as the mean of the terminal temperature differences between hot and cold fluid entering and leaving the heat exchanger, the difference being determined as follows:

$$\Delta Tm=\frac{(T_2-T_3)-(T_1-T_4)}{2.3\log\frac{(T_2-T_3)}{(T_1-T_4)}}$$

Metal, Metallic Element or Alloy	Temperature - t - (°C) (K) (°F)	Thermal Con - k - (W/m (Btu/(ft h
	527	371
"	727	357
"	927	342
Copper, electrolytic (ETP)	0 - 25	390
Copper - Admiralty Brass	20	111
Copper - Aluminum Bronze (95% Cu, 5% Al)	20	83
Copper - Bronze (75% Cu, 25% Sn)	20	26
Copper - Brass (Yellow Brass) (70% Cu, 30% Zn)	20	111
Copper - Cartridge brass (UNS C26000)	20	120
Copper - Constantan (60% Cu, 40% Ni)	20	22.7
Copper - German Silver (62% Cu, 15% Ni, 22% Zn)	20	24.9
Copper - Phosphor bronze (10% Sn, UNS C52400)	20	50
Copper - Red Brass (85% Cu, 9% Sn, 6%Zn)	20	61

Section III: Stoichiometry

1. $Ag(s)+H_2S(g)+O_2(g) \longrightarrow Ag_2S(s)+H_2O(l)$
2. $P_4(s)+O_2(g) \longrightarrow P_4O_{10}(s)$
3. $Pb(s)+H2O(l)+O2(g) \longrightarrow Pb(OH)2(s)$
4. $Fe(s)+H_2O(l) \longrightarrow Fe_3O_4(s)+H_2(g)$
5. $H_2(g)+I_2(s) \longrightarrow HI(s)$
6. $Fe(s)+O_2(g) \longrightarrow Fe_2O_3(s)$
7. $Na(s)+H_2O(l) \longrightarrow NaOH(aq)+H_2(g)$
8. $(NH_4)_2Cr_2O_7(s) \longrightarrow Cr_2O_3(s)+N_2(g)+H_2O(g)$
9. $P_4(s)+Cl_2(g) \longrightarrow PCl_3(l)$

Please <u>balance</u> the following Chemical equation **(2 points)**

$$Fe_2O_{3\ (s)} + CO_{(g)} \leftrightarrow Fe_{(s)} + CO_{2\ (g)}$$

Given that you have 20g of Iron (III) Oxide and 32g of Carbon Monoxide, find the <u>limiting reagent</u> and <u>how much Iron metal</u> you will yield, given the reaction proceeds to completion. **(3 points)**

Find the <u>volume</u> of Carbon Monoxide needed for it to be 32g, given the reaction is done at standard temperature and pressure.
(3 points)

Given the rate constant of the reaction, k, is 9.63e-3, and the reaction is first order, calculate the <u>half life</u> of the reaction. **(3 points)**

Bonus Question (5 points): Name one Nobel Laureate in Chemistry

Marie Curie